Financial text risk engine

Module A: sentiment-based mock index

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Problem and approach

Problem

News and social posts describe events that affect companies.

The text has no fixed format. A program must turn it into fields it can use.

Our approach

Read text from two sources.

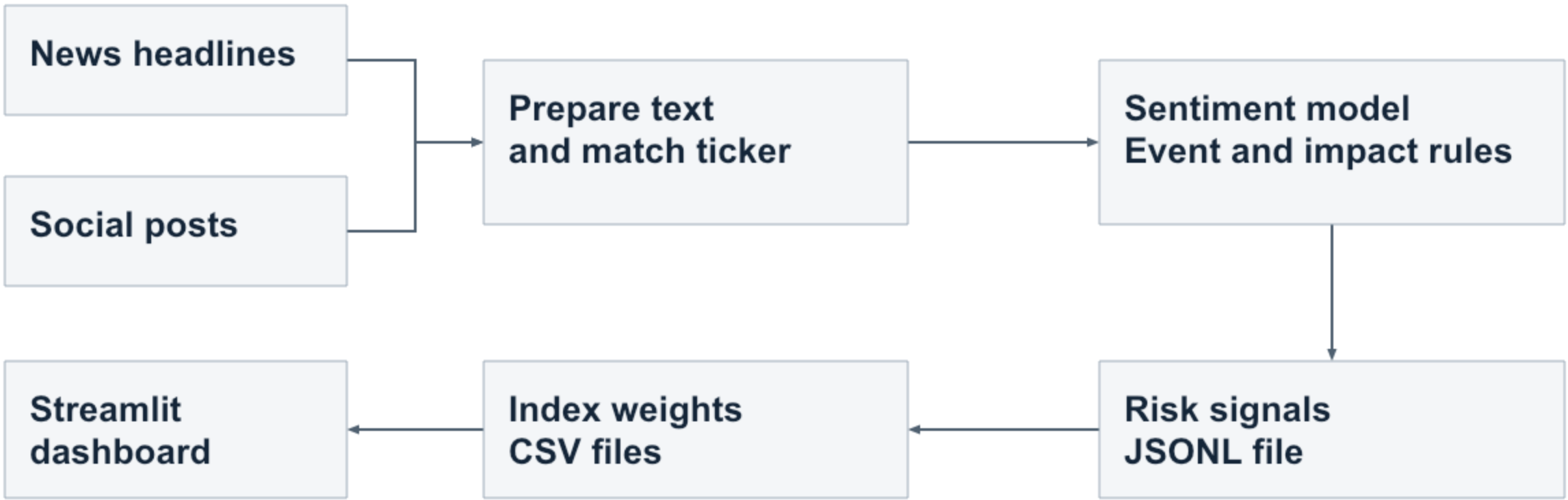
Produce sentiment, an event type and an impact score.

Use sentiment to change a mock index.

Output for each item

Company ticker Sentiment: −1 to +1 Event type Impact: 1 to 10

System design



The dashboard also accepts one new text item for a temporary simulation.

Implementation

Text and model

Historical financial posts and 10 source-linked news samples.

TF-IDF turns words into numbers. Logistic regression predicts bearish, bullish or neutral.

Events and weights

Keyword rules assign an event type. Fixed rules give an impact score.

The index starts with 10 stocks at 10% each. Only recognized events enter rebalancing.

Weight calculation

Desired change = 0.2 × starting weight × average sentiment

Transfer weight from negative to positive stocks. Keep the total at 100%.

Results

80.2%

Social sentiment accuracy

Macro F1: 0.744
Validation: 2,388 posts
Removed 68 overlapping training rows

Examples from the default sample run

Stock	Start (%)	Final (%)
AAPL	10.0000	10.1099
AMZN	10.0000	10.3241
META	10.0000	9.2598

24 input signals. 13 recognized events. Total final weight: 100%.

All 9 code tests pass. A clean checkout runs the full pipeline.

The accuracy above measures social sentiment only. We have not tested investment returns.

Use in financial analysis

Review company events

An analyst can filter news and social signals by source.

Each signal shows the text, company, score and event type. Source links help review the evidence.

Inspect index changes

The dashboard shows how sentiment changes the mock weights.

An analyst can enter a new headline and inspect its effect before changing saved data.

Current value

A repeatable way to inspect text signals and test a simple weighting rule.

Limitations and next steps

Current limits

News sentiment has no separate accuracy test.

Event labels and impact scores use rules.

Replay uses file order. Live feeds need a manual run.

Next steps

Label more news and test the model on it.

Test event labels and impact assumptions.

Add scheduled updates and real timestamps. Backtest weights with prices and trading costs.

Code and setup instructions

github.com/parth-chourasia/BITS_Goa-Parth_Chourasia_hackathon